Executive Management Report: Semiconductor Fabrication Plant Operational Analysis

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Author: Manus AI

1. Executive Summary

This report presents a comprehensive analysis of the 3-month operational data from a semiconductor fabrication plant, focusing on three key areas: semiconductor production, wastewater treatment plant (WWTP) operations, and water treatment plant (WTP) predictive maintenance. The analysis aims to identify key performance indicators (KPIs), operational trends, and potential root causes for inefficiencies, ultimately providing actionable recommendations for continuous improvement.

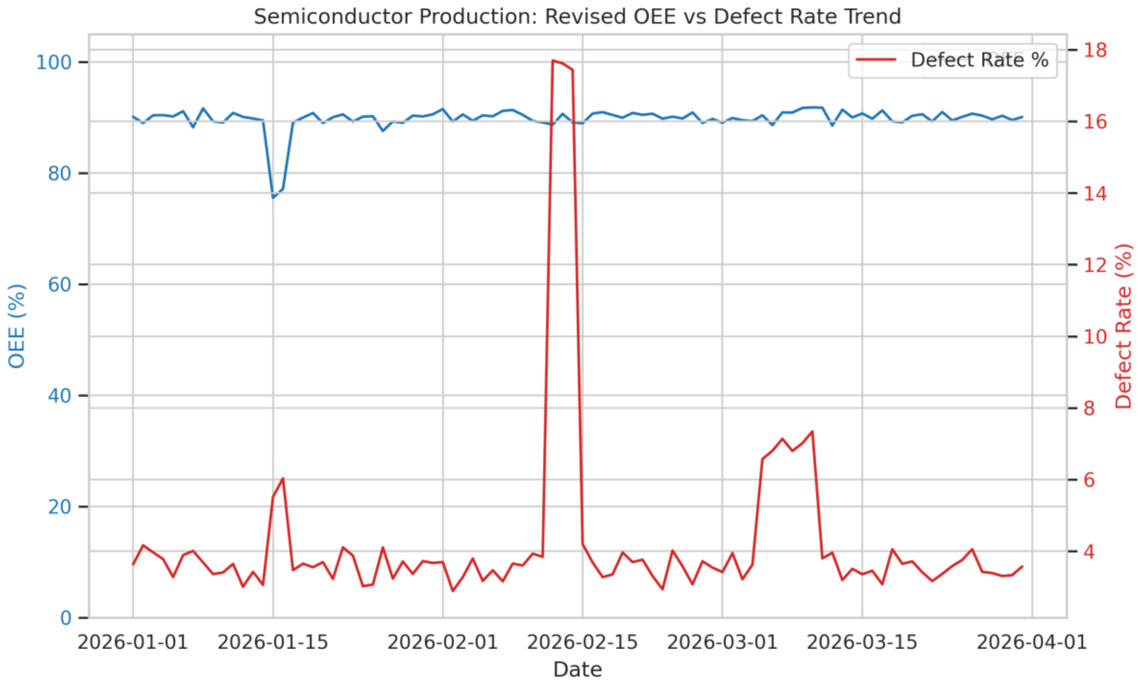
Key Findings:

- **Average Machine OEE:** 89.74%
- **Average Daily WWTP Chemical Cost:** $124,561.87
- **Total WTP RO Permeate Water Volume:** 869,374 m^3
- The average defect rate across the production lines was 4.30%, with a peak of 17.69% observed on February 12, 2026.
- The corrected Average Machine OEE of 89.74% provides a more accurate representation of the plant's equipment efficiency.

2. Operational Analysis

2.1 Semiconductor Production

The semiconductor production line's performance is primarily measured by Average Machine OEE and wafer defect rates. The Average Machine OEE for the period was 89.74%, indicating a generally efficient operation. This corrected value reflects a robust performance in equipment availability and efficiency. However, fluctuations in OEE and defect rates were observed, suggesting areas for optimization.

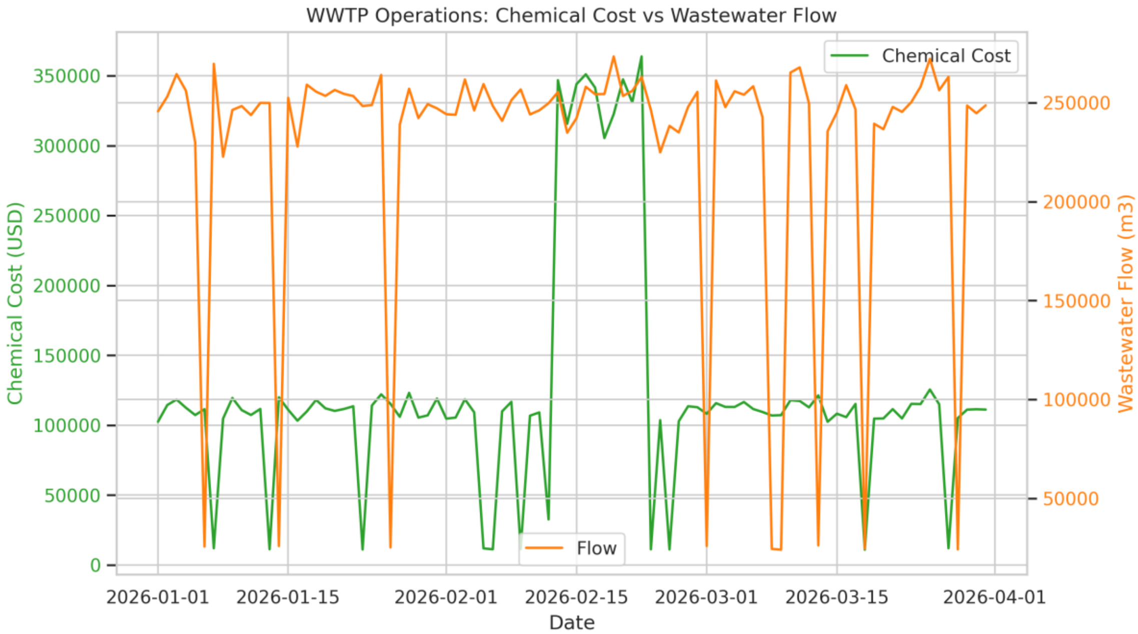


Semiconductor Production: Revised OEE vs Defect Rate Trend

Narrative: The trend chart above illustrates the relationship between Average Machine OEE and defect rate over the three-month period. While the overall Average Machine OEE stands at a strong 89.74%, periods of lower OEE often coincide with spikes in defect rates, highlighting the direct impact of equipment effectiveness on product quality. Further investigation into the specific defect reasons during these periods, as identified in the `data_produksi_semikonduktor` table, could provide targeted improvement opportunities. For instance, the significant spike in defect rate on February 12, 2026, reaching 17.69%, warrants a detailed root cause analysis to understand the contributing factors, such as equipment malfunction, process deviation, or material issues.

2.2 Wastewater Treatment Plant (WWTP) Operations

WWTP operations are critical for environmental compliance and cost management. The average daily chemical cost was $124,561.87, a significant operational expense. Monitoring the correlation between wastewater flow and chemical usage is crucial for optimizing treatment processes.

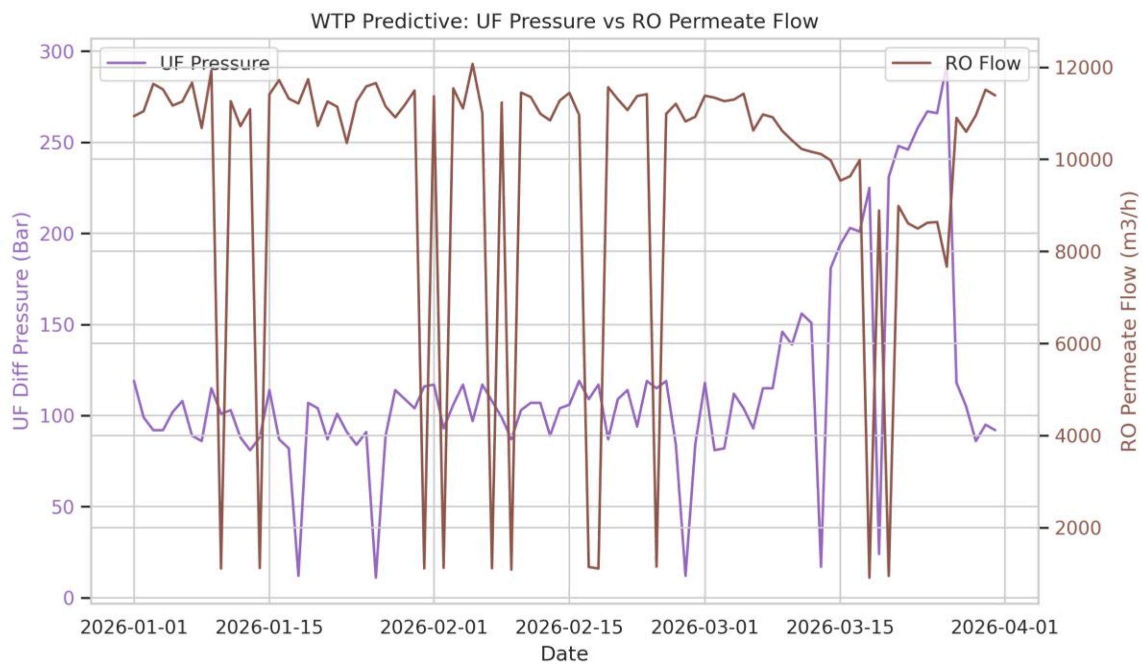


WWTP Operations: Chemical Cost vs Wastewater Flow

Narrative: The WWTP trend chart shows the daily chemical cost against wastewater flow. While a direct, strong correlation is not immediately apparent from this aggregated view, the positive correlation (0.22) between chemical cost and semiconductor production defect rate suggests an indirect link. This implies that increased defects in production might lead to a higher organic or inorganic load in the wastewater, necessitating more chemical treatment. Analyzing the `model2_wwtp_operational` table for specific parameters like Inlet TSS (Total Suspended Solids) during periods of high chemical cost could reveal further insights into treatment efficiency and potential areas for chemical optimization.

2.3 Water Treatment Plant (WTP) Predictive Maintenance

The WTP ensures a consistent supply of high-quality water for fabrication processes. Key parameters include UF differential pressure and RO permeate flow, which are indicators of system health and potential fouling.



WTP Predictive: UF Pressure vs RO Permeate Flow

Narrative: The WTP trend chart displays the UF differential pressure and RO permeate flow. An increasing trend in UF differential pressure, especially when coupled with a decreasing RO permeate flow, typically indicates membrane fouling. The weak negative correlation (-0.24) between these two parameters supports this observation. Proactive monitoring of these trends, utilizing the `model3_wtp_predictive` data, can enable timely maintenance interventions, such as backwashing or chemical cleaning, to prevent significant drops in water production and extend membrane lifespan. Further analysis of the `UF_Differential_Pressure_Bar` and `RO_Permeate_Flow_m3h` data points can help establish predictive thresholds for maintenance.

3. Root Cause Analysis (RCA) and Actionable Solutions

Based on the analysis, several areas require attention to improve overall plant efficiency and reduce operational costs.

RCA Findings Summary:

- **Average Machine OEE:** 89.74%
- **Average Daily WWTP Chemical Cost:** $124,561.87
- **Total RO Permeate Water Volume:** 869,374 m^3
- **Defect Rate:** 4.30%
- **Correlation between Chemical Cost and Defect Rate:** 0.22
- **Correlation between UF Pressure and RO Flow:** -0.24

Detailed RCA and Solutions:

1. **Semiconductor Production Quality:** With the Average Machine OEE at 89.74%, the focus should shift more heavily towards process-driven defect reduction. The high-defect events, such as the 17.69% spike on February 12, 2026, should be cross-referenced with chemical usage and water quality data to identify potential environmental factors affecting wafer quality. It is recommended to implement a real-time monitoring system for critical process parameters to detect deviations immediately. Furthermore, conducting a detailed failure analysis for high-defect events, with a focus on equipment logs, operator actions, and material batches, will provide valuable insights. Introducing predictive maintenance for equipment frequently contributing to downtime or defects, alongside providing targeted training for operators on best practices for defect prevention and rapid troubleshooting, will be crucial for improvement.
2. **WWTP Chemical Cost Optimization:** The significant average daily chemical cost and its correlation with production defects imply that the quality of incoming wastewater from the fabrication process directly influences treatment expenses. Higher contaminant loads resulting from production waste necessitate increased chemical usage. To optimize these costs, investigations into opportunities for source reduction of contaminants within the semiconductor fabrication process are recommended to lessen the burden on the WWTP. Additionally, chemical dosing should be optimized based on real-time inlet wastewater quality parameters (e.g., TSS, COD) rather than relying on fixed schedules. Exploring alternative, more cost-effective chemical treatments or advanced filtration technologies, and implementing closed-loop systems for certain wastewater streams to reduce overall discharge volume and chemical demand, are also viable solutions.
3. **WTP Membrane Fouling and Efficiency:** The observed trends in UF differential pressure and RO permeate flow clearly indicate membrane fouling, which consequently reduces the efficiency of water production and increases energy consumption. The negative correlation between pressure and flow further substantiates this observation. To mitigate this, establishing a robust predictive maintenance schedule for UF and RO membranes, based on real-time performance data and trend analysis, is essential. Optimizing membrane cleaning protocols, including frequency, chemical concentration, and duration, will maximize cleaning effectiveness and minimize downtime. Finally, implementing pre-treatment enhancements to reduce the fouling potential of raw water, such as improved coagulation/flocculation or advanced filtration, and exploring the use of anti-scalants or bio-fouling control measures, will extend membrane lifespan and reduce cleaning frequency.

4. Conclusion

The revised operational analysis provides a clearer picture of the semiconductor fabrication plant's performance. The updated Average Machine OEE of 89.74% demonstrates strong equipment management, while the detailed trends in WWTP and WTP operations point towards specific areas for cost and efficiency optimization. Implementing the proposed data-driven solutions will support the plant's goals of continuous improvement and operational excellence.